

cidence angle. The calculated scattering spectra of a heptamer for different polarization angles are shown in Fig. 3C, where the cluster geometry is identical to that used in Fig. 3A. These spectra display Fano minima at 1450 nm, with asymmetric line shapes that match the experimental spectra. The nanoshell separation modeled here is smaller than that used for the trimer calculations to account for the strongly red-shifted Fano minimum. This red shift is probably due to a combination of at least three factors: (i) smaller nanoshell separation due to inhomogeneous self-assembled monolayer coverage, (ii) a higher-refractive-index environment near the cluster due to excess polymer deposition, and (iii) increased capacitive coupling between the nanoparticles due to nanoshell faceting.

This cluster concept can be generalized to other functional 2D and 3D structures. One example is the tetrahedral cluster, which supports isotropic electric and magnetic resonances in three dimensions (28) and can be used as a building block for isotropic metamaterials. Symmetry breaking can be applied to engineer other types of optical modes: Trimers comprising three different particle types support magnetoelectric modes, and tetrahedral clusters comprising four different particle types are chiral. Nonspherical plasmonic particles can be used to construct more elaborate structures, provided that their orientations can be controlled during assembly. In all of these structures, resonances can be tuned by varying individual particle geometries, interparticle separation, and the dielectric environment of the cluster. The assembly of clusters from solution is highly versatile: It can lead to liquid metamaterials or “metafluids” (28), be integrated into soft materials such as gels, or be encapsu-

lated and dried onto surfaces of arbitrary curvature or patterning. Future work will focus on these applications and on achieving higher cluster yields comparable to those attained with lithographically defined patterns (29), emulsion droplets (30), and DNA linking (31).

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How Grain Growth Stops: A Mechanism for Grain-Growth Stagnation in Pure Materials

Elizabeth A. Holm* and Stephen M. Foiles

The thermodynamic equilibrium state of crystalline materials is a single crystal; however, polycrystalline grain growth almost always stops before this state is reached. Although typically attributed to solute drag, grain-growth stagnation occurs, even in high-purity materials. Recent studies indicate that grain boundaries undergo thermal roughening associated with an abrupt mobility change, so that at typical annealing temperatures, polycrystals will contain both smooth (slow) and rough (fast) boundaries. Mesoscale grain-growth models, validated by large-scale polycrystalline molecular dynamics simulations, show that even small fractions of smooth, slow boundaries can stop grain growth. We conclude that grain-boundary roughening provides an alternate stagnation mechanism that applies even to high-purity materials.

Most metals and ceramics are polycrystalline: They are made up of many individual crystallites, called grains, separated by internal interfaces or grain boundaries. When polycrystalline materials are annealed

at sufficiently high temperatures, grain boundaries move and rearrange so as to increase the average grain size and decrease the grain-boundary area per unit volume. However, even at very high temperatures, grain growth only

rarely proceeds to the equilibrium single-crystal state. Instead, grain growth usually stops, though substantial internal interface remains. In fact, grain stagnation is so pervasive that most grain-growth models presume a finite maximum grain size based purely on empirical observations (1).

Understanding and controlling grain growth is important to nearly every engineered material. For materials that rely on strength, toughness or formability, including most nanocrystalline materials, a stable, fine grain size is desirable. However, there are also important systems, such as superalloy turbine blades and silicon photovoltaics, in which a large (or even single-crystal) grain size is preferred.

There have been many grain-growth stagnation mechanisms proposed, each valid in certain regimes. Well-known processes that reduce the driving force for grain growth sufficiently to

Computational Materials Science and Engineering Department, Sandia National Laboratories, Albuquerque, NM 87185–1411, USA.

*To whom correspondence should be addressed. E-mail: eaholm@sandia.gov

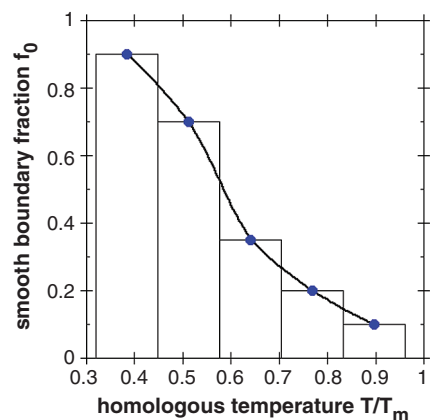


Fig. 1. Distribution of grain-boundary roughening-transition temperatures for several hundred grain boundaries in Ni calculated by MD simulations (17). The width of the bars corresponds to the temperature bin width; data points are located at the mean temperature of each bin. The solid line is a guide for the eye.

produce a metastable polycrystal include pinning by dispersed particles (2), pinning by thermal grooves (3), and the film-thickness effect (4). These occur only in films or in systems containing an immobile second phase. In other materials, grain-growth stagnation is often attributed to the segregation of solute species to the grain boundary. Solute atoms inhibit grain growth by requiring the boundary to move in concert with its solute cloud (5–9) or by directly changing boundary-migration mechanisms (10, 11). For most solutes and temperatures of interest, this substantially slows grain-boundary motion and, hence, also slows grain growth. However, grain-growth stagnation is also observed in systems in which solutes should not limit growth, including high-purity materials (12, 13), nanocrystalline materials that should be in the solute breakaway regime (14, 15), and materials at high homologous temperatures where solute diffusivity should be high (15). It is reasonable to suppose that there might be another active grain-growth stagnation mechanism in these systems.

We recently calculated grain-boundary mobility, which scales how fast a boundary can move in response to a driving force, for a catalog of 388 grain boundaries in Ni using a synthetic-driving force molecular dynamics (MD) method (16, 17). At a given temperature, mobility falls into two ranges: (i) high-mobility (fast) boundaries and (ii) low-mobility (slow) boundaries, whose mobility is too small for us to measure via MD. Typical high-mobility boundaries move continuously with well-defined activation energies and atomically rough-boundary structures. On the other hand, low-mobility boundaries move in a jerky, stepwise manner with different activation energies and atomically smooth boundary structures (18). The boundary structure changes from smooth/low-mobility to rough/high-mobility at a characteristic temperature, T_r

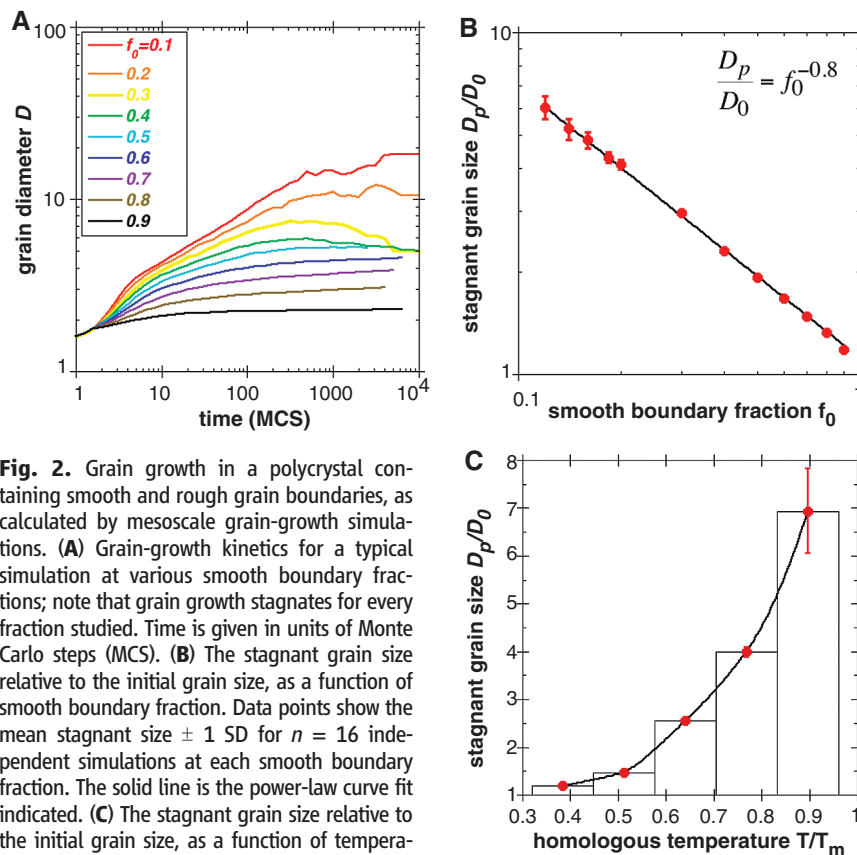


Fig. 2. Grain growth in a polycrystal containing smooth and rough grain boundaries, as calculated by mesoscale grain-growth simulations. (A) Grain-growth kinetics for a typical simulation at various smooth boundary fractions; note that grain growth stagnates for every fraction studied. Time is given in units of Monte Carlo steps (MCS). (B) The stagnant grain size relative to the initial grain size, as a function of smooth boundary fraction. Data points show the mean stagnant size ± 1 SD for $n = 16$ independent simulations at each smooth boundary fraction. The solid line is the power-law curve fit indicated. (C) The stagnant grain size relative to the initial grain size, as a function of temperature. Bar width is from Fig. 1, and data points and error bars are extracted from the data in Fig. 2B. The solid line is a guide for the eye.

(the roughening temperature), which can vary by hundreds of degrees from boundary to boundary, as discussed by Olmsted *et al.* (17). Figure 1 shows the distribution of roughening temperatures for our catalog of Ni boundaries: At low temperature, most boundaries are smooth and slow, but even at $0.9 T_m$ (where T_m is the melting temperature), $\sim 10\%$ of the boundaries remain smooth and slow.

It is important to note that the roughening transition is not related to faceting, in which certain crystallographically favored grain boundaries decompose into flat facets that lie along low-energy inclinations (19–21). Because faceting results in both high and low grain-boundary mobilities (22, 23), it is not generally implicated in grain-growth stagnation (24). In contrast, the roughening transition occurs in all boundaries, not just those that are thermodynamically inclined to facet, and the structure of smooth boundaries does not include crystallographic facets (18). Smooth boundaries generally have very low mobility, which could inhibit grain growth. Because faceting and roughening are distinct phenomena, it is possible that a single grain boundary could undergo both transitions at different temperatures; however, this report concerns only the microstructural effects of grain-boundary roughening and its associated mobility transition.

To determine the effect of boundary roughening on grain growth in a polycrystal, we incorporate the boundary mobility and roughening data in

Fig. 1 into a mesoscale simulation of polycrystalline microstructural evolution (25–28). When all boundaries are rough/high-mobility, we find that grain growth proceeds to the equilibrium single-crystal state. In contrast, in simulations that include smooth/low-mobility boundaries, grain growth slows dramatically at a finite grain size that decreases as the fraction of smooth boundaries, f_0 , increases, as shown in Fig. 2A. It is not surprising that immobile boundaries can slow or stop grain growth; however, it is unexpected that the grain structure stagnates at such small smooth boundary fractions, because the motion of mobile boundaries can sweep out immobile boundaries. Our results show that in a polycrystal, grain boundaries form an interconnected network that may be pinned, even when many individual boundaries remain unpinned. This observation applies, not only to pinning by smooth boundaries, but also to other stagnation mechanisms. For example, particle pinning theories disagree on whether all or merely a fraction of boundaries must be pinned by intersecting particles for the grain structure to be stabilized (29, 30). The present results indicate that pinning a fraction of boundaries is sufficient.

The stagnant grain size relative to the initial grain size (that is, D_p/D_0) decreases as a power law in f_0 , as shown in Fig. 2B. By combining Fig. 1 and Fig. 2B, we can determine the stagnant grain size as a function of temperature, as shown

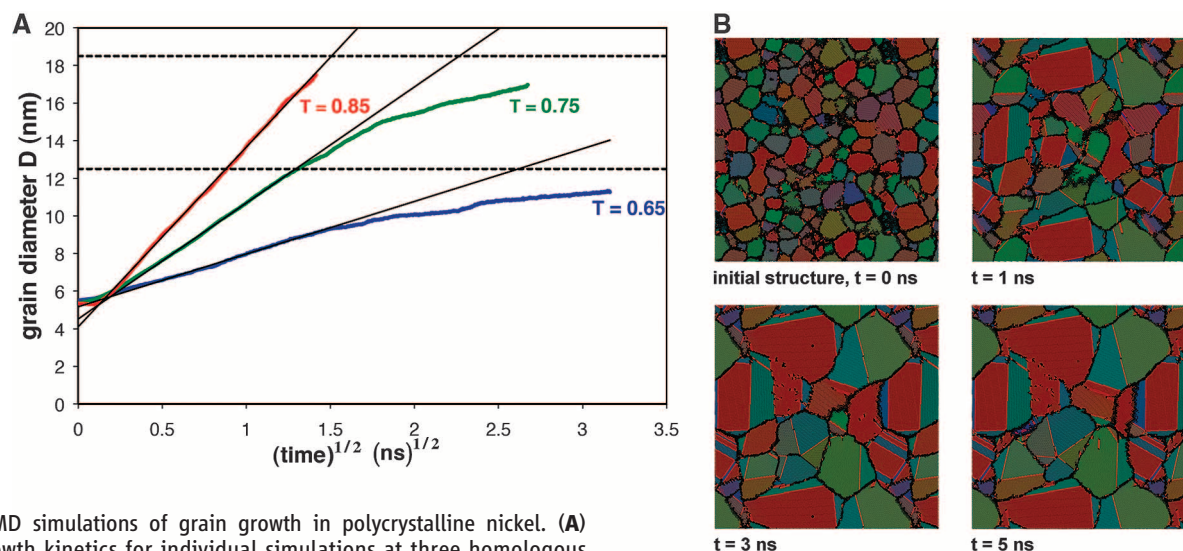


Fig. 3. MD simulations of grain growth in polycrystalline nickel. **(A)** Grain-growth kinetics for individual simulations at three homologous temperatures. Solid black lines show normal growth kinetics with grain diameter proportional to the square root of time; dashed lines indicate the stagnant grain size predicted by mesoscale simulations. **(B)** Time evolution of a slice through the 3D system at $T = 0.75 T_m$ at

0, 1, 3, and 5 ns. The atomic color scheme reflects local lattice orientation: Red atoms are hexagonal close-packed coordinated (i.e., dislocations, twins); black atoms are of unknown coordination (i.e., grain boundaries).

in Fig. 2C. At low temperatures, very little grain growth can occur due to a large fraction of smooth/low-mobility boundaries. Even at $0.9 T_m$, which is higher than many industrial-annealing temperatures, the grain size can increase only by one order of magnitude before being pinned by smooth boundaries. Within our mesoscale simulations, smooth boundaries can provide an effective grain-growth pinning mechanism, but are they likely to be a factor in real materials?

Although we observed grain-boundary roughening in flat grain boundaries (18), boundaries in real polycrystals are curved. It seems reasonable to wonder whether the geometric constraints imposed by boundary curvature might prevent the realization of a smooth/low-mobility boundary structure. The obvious solution is to measure the atomic roughness of curved boundaries directly, but this poses a challenge for boundaries of unknown shape. Furthermore, geometric roughness is an inherently noisy measurement, and for small boundaries, the noise often obscures the signal of the roughening transition (18). Mobility is a reliable indicator of boundary roughness, but it is not feasible to measure the mobility of boundaries in a polycrystal, in which the driving force is poorly characterized and continually changing. Although we do not yet have a robust metric to probe boundary roughness in a polycrystal, we can look for the effects of smooth boundaries indirectly via atomic-scale grain-growth simulations.

We begin by building realistic, three-dimensional (3D) nickel polycrystals. Grains are randomly oriented, and grain boundaries assume the structure, crystallography, energy, and mobility naturally imposed by the orientations of the two neighboring grains. We used MD to evolve these structures at various temperatures (28, 31, 32). Figure 3A shows the evolution of grain size with

time. After an initial transient, the average grain radius increases with the square root of time, as expected for normal grain growth (4). Some investigators have suggested that, in nanocrystals, grain growth should be governed by triple junction mobility and should follow a linear growth law (33), and a previous polycrystal MD simulation seemed to support this (34). However, those simulations examined only the first 500 ps of growth. Our results show that, after a short initial transient, nanocrystalline grain growth follows the same $t^{1/2}$ kinetics (where t is time) observed in microcrystalline materials.

After a period of normal grain growth, Fig. 3A shows that grain growth slows dramatically at the two lower temperatures that we studied. Earlier simulations of grain growth in a perfectly pure metal were not able to access the system size or time scale necessary to reach the grain size where slowing occurs (34). A cross section of a stagnant polycrystal is shown in Fig. 3B; there are no apparent microstructural features that distinguish this stagnant microstructure from a freely evolving one.

The mesoscale grain-growth simulation data in Fig. 2C suggest that smooth grain boundaries should pin grain growth at a relative grain size $D_p/D_0 = 2.5$ (12.5 nm) at $T = 0.65 T_m$ (where T is temperature) and at $D_p/D_0 = 3.7$ (18.5 nm) at $T = 0.75 T_m$. [At $T = 0.85 T_m$, the pinned relative grain size $D_p/D_0 = 5.6$ (28 nm) is beyond the limitations of the MD system size.] These pinned grain sizes are shown as dotted lines in Fig. 3A. The stagnant grain sizes predicted by the mesoscale grain-growth model are in agreement with the stagnant grain sizes observed in the MD simulations. Although this is not conclusive proof that smooth boundaries are a mechanism of grain-growth stagnation in pure polycrystals, it is highly suggestive.

Controlling grain size to optimize properties is a fundamental problem in materials science. One aspect of this is understanding how grain growth stops. A number of mechanisms have been proposed to explain various instances of grain-growth stagnation. Smooth-boundary pinning offers a mechanism for how grain growth stops in perfectly pure materials, even at high homologous temperatures.

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Interface-Confined Ferrous Centers for Catalytic Oxidation

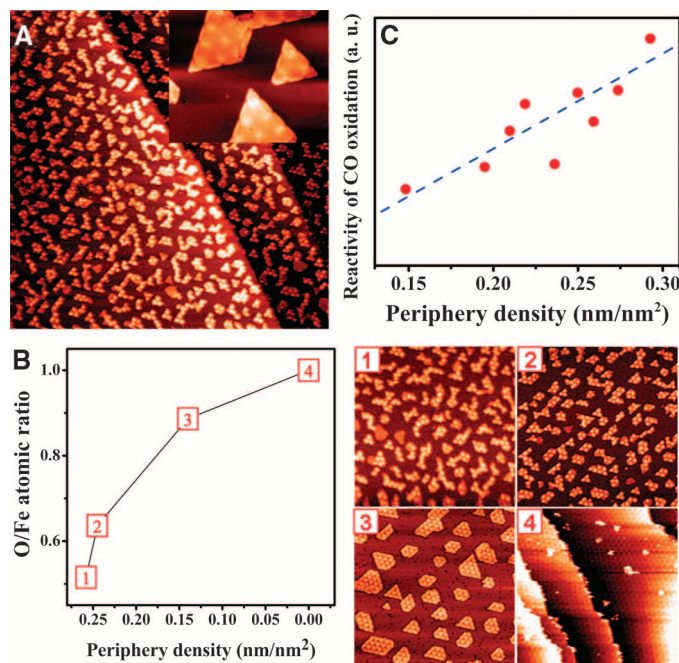
Qiang Fu,^{1*} Wei-Xue Li,^{1,2*} Yunxi Yao,¹ Hongyang Liu,¹ Hai-Yan Su,^{1,2} Ding Ma,¹ Xiang-Kui Gu,^{1,2} Limin Chen,¹ Zhen Wang,¹ Hui Zhang,³ Bing Wang,³ Xinhe Bao^{1†}

Coordinatively unsaturated ferrous (CUF) sites confined in nanosized matrices are active centers in a wide range of enzyme and homogeneous catalytic reactions. Preparation of the analogous active sites at supported catalysts is of great importance in heterogeneous catalysis but remains a challenge. On the basis of surface science measurements and density functional calculations, we show that the interface confinement effect can be used to stabilize the CUF sites by taking advantage of strong adhesion between ferrous oxides and metal substrates. The interface-confined CUF sites together with the metal supports are active for dioxygen activation, producing reactive dissociated oxygen atoms. We show that the structural ensemble was highly efficient for carbon monoxide oxidation at low temperature under typical operating conditions of a proton-exchange membrane fuel cell.

In many catalytic processes, the size of metal-containing catalysts falls typically in the range of 1 to 10 nm. The catalytically active sites in these catalysts are often coordinatively unsaturated metal cations that are able to undergo facile electron transfer and promote catalytic reactions, as has been studied in (1–5). In oxygen-carrying hemoglobin, the iron (II) ion constrained in a planar porphyrin ring could transport O₂ via facile interconversion between Fe²⁺ and Fe³⁺ states (6). Homogeneous catalysis in aqueous solutions uses Fenton's reagent, being a mixture of Fe²⁺ ions and H₂O₂, to produce hydrated ferryl ions [(H₂O)₅Fe^{IV}O]²⁺, which act as the active intermediates to oxidize various organic compounds, such as methane (7). However, for heterogeneous catalysis with nanopore environments the active species in Fe-ZSM-5 and Fe-silicalite zeolites are also coordinatively unsaturated Fe²⁺ grafted to the zeolite crystalline matrix. The dissociation of N₂O at the Fe sites leads to formation of “ α -oxygen” species, which are active in the selective oxidation of benzene to phenol at mild con-

ditions (8). These characteristic coordinatively unsaturated ferrous (CUF) sites are confined by various ensembles such as proteins, ligands, and nanopore matrix, which are essential for their high reactivity and stability in the catalytic oxidation reactions mentioned above (9).

Fig. 1. (A) STM image (200 nm × 200 nm) from the 0.25-ML FeO_{1-x}/Pt(111) surface. (Inset) An atomic-resolution STM image of FeO monolayer nanoislands (25 nm × 20.8 nm) recorded at liquid N₂ temperature. **(B)** Ratios of XPS O 1s to Fe 2p_{3/2} peak intensity from FeO_{1-x}/Pt(111) surfaces with different periphery density of FeO nanoisland. Samples 1 to 3 are 0.25-ML FeO nanoislands prepared at 1.3 × 10⁻⁶ mbar O₂ and annealed in UHV at 473, 573, and 673 K, respectively. Sample 4 is the full-monolayer FeO film on Pt(111). The STM images are all 100 nm by 100 nm. **(C)** Dependence of reactivity of CO oxidation on the periphery density at 0.25-ML FeO_{1-x}/Pt(111) surfaces.



¹State Key Laboratory of Catalysis, Dalian Institute of Chemical Physics, Chinese Academy of Sciences, Dalian 116023, China.
²Center for Theoretical and Computational Chemistry, Dalian Institute of Chemical Physics, Chinese Academy of Sciences, Dalian 116023, China.
³Hefei National Laboratory for Physical Sciences at the Microscale, University of Science and Technology of China, Hefei 230026, China.

*These authors contributed equally to this work.

†To whom correspondence should be addressed. E-mail: xhbao@dicp.ac.cn



How Grain Growth Stops: A Mechanism for Grain-Growth Stagnation in Pure Materials

Elizabeth A. Holm and Stephen M. Foiles

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Taking the Rough with the Smooth

Even with extensive annealing at high temperatures, most polycrystalline materials will not become a perfect single crystal, which would represent the thermodynamically preferred state. The stability of the polycrystalline state has been attributed to the presence of impurities that accumulate at the grain boundaries, but even very pure materials show grain growth stagnation. Using simulations, **Holm and Foiles** (p. 1138) show that grain boundaries can be classified as “rough” and “smooth.” Rough boundaries move continuously with well-defined activation energies, while the smooth boundaries have low mobility and move in a jerky, stepwise manner. With heating, a boundary can change from smooth to rough, but the transition temperature can vary by hundreds of degrees from one grain boundary to the next. These smooth, low-mobility boundaries thus pin the polycrystalline structure, even in the absence of impurities.

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